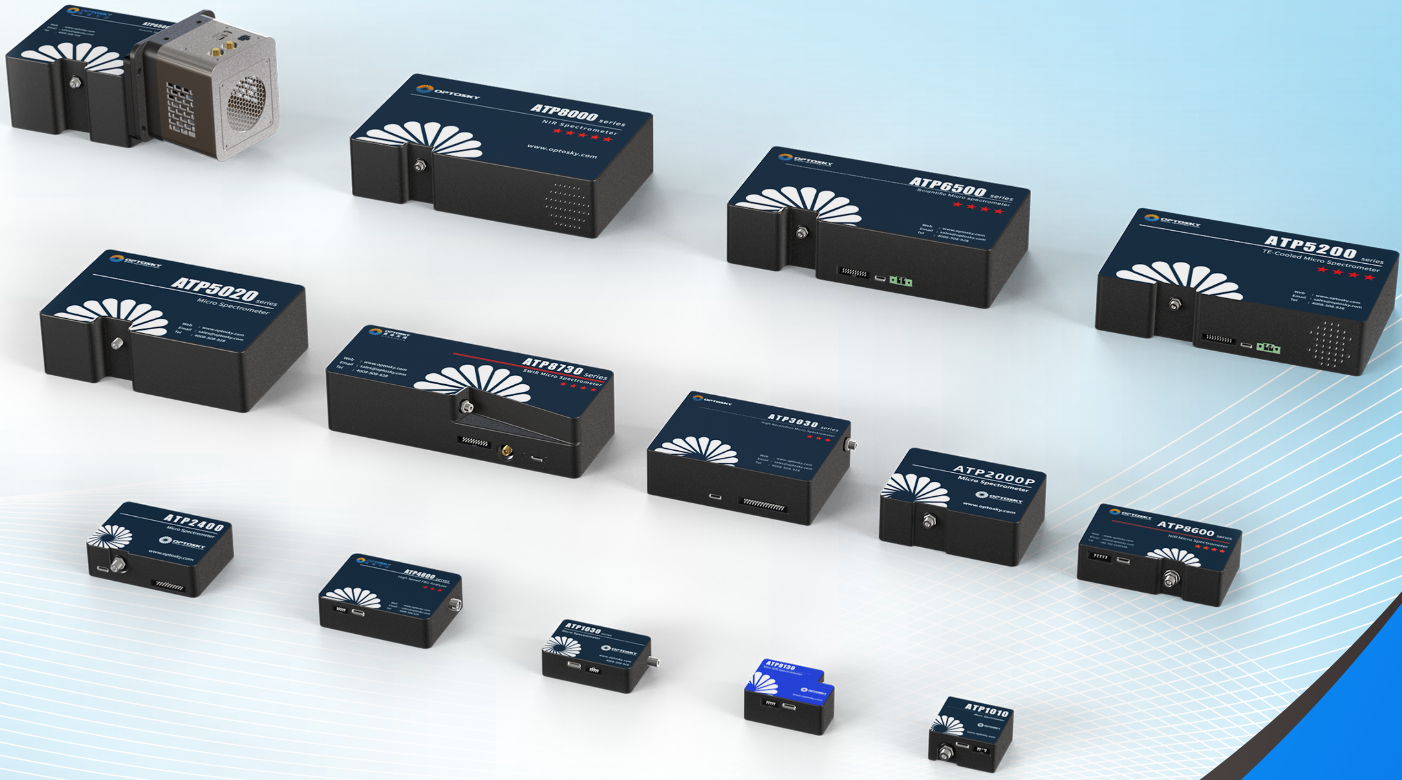
OPTOSKY
Spectroscopy Solutions

20-YEAR
SPECTROSCOPY MANUFACTURER



ATP SERIES FIBER OPTIC SPECTROMETER

Communication Protocol

Contents

1. Electric Parameters

Tab. 1.1 Electrical Characteristics

Variable	Min	Typ	Max	Unit
Power Supply				
Operating Voltage Range	4.5	5	5.5	V
Operating Current		170		mA
Logic Input (3.3V LVTTL, 5V Tolerant)				
High-Level Input Voltage (VIH)	1.7		3.6	V
Low-Level Input Voltage (VIL)	-0.3		1.0	V
Logic Output (3.3V LVTTL)				
High-Level Output Voltage (VOH)	2.4			V
Low-Level Output Voltage (VOL)			0.4	V

2. ATP Series Spectrometer General Command Communication Protocol

2.1. Serial Port Communication Parameters

Tab. 2.1 Serial Interface Defaults

Variable	Default Value
Data Rate (Baud)	115200
Data Bits	8
Parity	None
Stop Bits	1
Flow Control	None

2.2. General Command Protocol and Format

All data sent or received is encoded in **Hexadecimal** format.

Data Position	0 1	2 3	4	5 ... N-1	N
Chinese Description	<帧头> (2 字节)	<数据长度> (2 字节)	<命令> (1 字节)	<有效数据> (数据长度-4 个字节)	<校验位> (1 字节)
English Description	<Head> (2bytes)	<Length> (2bytes)	<Cmd> (1bytes)	<Data> ((Length-4)bytes)	<Checksum> (1bytes)

Header:

Used to locate the start of the communication packet, fixed as "AA 55".

Length:

The number of bytes from the "Length" field to the "Checksum" field.

Command (Cmd):

Command ID.

Data:

Determined by the command; the Data field can be empty. If the corresponding command in the Command Overview section has no <Data>, it indicates it is empty.

Checksum:

The sum of all bytes from the "Length" field to the "Data" field, taking the lower 8 bits of the result.

Checksum = (Length + Cmd + Data) & 0xFF

2.3. Command Overview

Tab. 2.2 General Command Preview

Cmd ID	Function Name	Cmd ID	Function Name
0x03	Get PN Number	0x43	Get Minimum Integration Time
0x04	Get SN Number	0x61	Set GPIO Pin Output Level
0x06	Get Module Production Date	0x0E	Restore Factory Settings
0x09	Get Module Manufacturer Info	0x14	Set Integration Time
0x0A	Get Device Pixel Length	0x16	Start CCD Scan (Asynchronous Mode)
0x1C	Get Slit Size	0x23	Acquire Dark Current (Asynchronous Mode)
0x46	Get Device Attributes	0x17	Read CCD Acquisition Data
0xAF	Get Firmware Version	0x1E	Start CCD Scan (Synchronous Mode)
0x0B	Set Serial Baud Rate	0x2F	Acquire Dark Current (Synchronous Mode)
0x0C	Get Serial Baud Rate	0x1F	Enable External Trigger Acquisition
0x12	Set TEC Temperature	0xF1	Disable External Trigger Acquisition
0x13	Get TEC Temperature	0x55	Get Wavelength Calibration Coefficients
0x28	Set Acquisition Average Count	0x24	Set Xenon Lamp Delay Trigger Time
0x2B	Get Acquisition Average Count	0xF2	Set Xenon Lamp Max Flash Count
0x41	Get Current Integration Time	0xF3	Set Xenon Lamp Frequency
0x42	Get Maximum Integration Time	0xF5	Get Xenon Lamp Attributes

2.3.1. System Parameter Functions

System parameters are set at the factory and cannot be modified by the user; they can only be read.

1） Get PN Number Cmd:03

Gets the module's PN (Part Number). Host sends empty Data field; Slave returns Data field of 11 bytes, representing the ASCII code of the module's PN.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	03	（Empty）	07
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 0F	03	41 54 50 31 30 31 30 30 30 30 30	79

Slave returns 11 bytes of valid data: 41 54 50 31 30 31 30 30 30 30 30, which converts to the ASCII string "ATP10100000".

2） Get SN Number Cmd:04

Gets the module's SN (Serial Number). Host sends empty Data field; Slave returns Data field of 16 bytes, representing the ASCII code of the module's SN.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	04	（Empty）	08
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 14	04	32 35 31 32 31 58 33 31 41 42 31 32 33 36 32 33	83

Slave returns 16 bytes of valid data: 32 35 31 32 31 58 33 31 41 42 31 32 33 36 32 33, which converts to the ASCII string "25121X31AB123623".

3） Get Module Production Date Cmd:06

Gets the module's production date. Host sends empty Data field; Slave returns Data field of 8 bytes, representing the ASCII code of the module's production date.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	06	（Empty）	0A
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 0C	06	32 30 32 35 30 33 32 31	A1

Slave returns 8 bytes of valid data: 32 30 32 35 30 33 32 31, which converts to the ASCII string "20250321", meaning the module production date is March 21, 2025.

4） Get Module Manufacturer Info Cmd:09

Gets the module manufacturer information. Host sends empty Data field; Slave returns Data field of 7 bytes, representing the ASCII code of the module manufacturer info, fixed as 4F 50 54 4F 53 4B 59, i.e., "OPTOSKY".

Example:

Host Send	Head	Length	Cmd	Data	Checksum

	AA 55	00 04	09	（Empty）	0D
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 0B	09	4F 50 54 4F 53 4B 59	0D

5） Get Device Pixel Length Cmd:0A

Gets the device's pixel length. Host sends empty Data field; Slave returns Data field of 2 bytes, representing the pixel length in decimal, high byte first, low byte last.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	0A	（Empty）	0E
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	0A	10 00	20

Slave returns 2 bytes of valid data: 10 00, which converts to decimal 4096.

6） Get Slit Size Cmd:1C

Gets the device's slit size, in μm. Host sends empty Data field; Slave returns Data field of 2 bytes, representing the slit size in decimal, high byte first, low byte last.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	1C	（Empty）	20
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	1C	00 19	3B

Slave returns 2 bytes of valid data: 00 19, which converts to decimal 25, indicating the device slit size is 25μm.

7） Get Device Attributes Cmd:46

Gets the device attributes. Host sends empty Data field; Slave returns Data field of 4 bytes.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	46	（Empty）	4A
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 08	46	10 00 00 07	65

The slave returns 4 bytes of valid data: 10 00 00 07.

The first two bytes (10 00) convert to decimal (4096) representing the device pixel length.

The third byte (00) is reserved, currently meaningless.

The fourth byte (07) converted to binary (0111):

Bit	Corresponding Value
Bit[0]	0：Integration time is 2 bytes 1：Integration time is 4 bytes
Bit[1]	0：Integration time unit is ms 1：Integration time unit is μs
Bit[2]	0：USB commands require checksum 1：USB commands do not require checksum
Bit[3]	Reserved

0111 means: Integration time 2 bytes (0), Integration time unit μs (1), USB commands do not require checksum (1).

8） Get Firmware Version Cmd:AF

Gets the firmware version number. The version number format is: V + major version + . + medium version + . + minor version. Host sends empty Data field; Slave returns Data field whose byte count depends on the version length, representing the ASCII code of the firmware version.

Example:

	Head	Length	Cmd	Data	Checksum
Host Send	AA 55	00 04	AF	（Empty）	B3
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 0A	AF	56 32 2E 32 2E 31	00

Slave returns 6 bytes of valid data: 56 32 2E 32 2E 31, which converts to the ASCII string "V2.2.1".

2.3.2. User Parameter Functions

1） Set Serial Baud Rate Cmd:0B

Changes the communication serial port baud rate. Takes effect after power cycle. If the user forgets the set baud rate, they can use the USB connection and the provided USB debugging software, send the Get Current Serial Baud Rate command (0C) to obtain the current baud rate, and can send the Restore Factory Settings command (0E) or the Set Serial Baud Rate command (0B) to modify the current baud rate.

Host sends Data field of 1 byte, representing the set serial baud rate. Range from 01 to 10.

01:9600 bps;
02: 19200 bps;
03: 19200 bps;
04: 38400 bps;
05: 56000 bps;
06: 57600 bps;
07: 115200 bps（Default）;
08: 128000 bps;
09: 230400 bps;
0A: 256000 bps;
0B: 460800 bps;
0C: 500000 bps;
0D: 512000 bps;
0E: 600000 bps;
0F: 750000 bps;
10: 921600 bps;

Slave returns Data field of 2 bytes.

Byte	Corresponding Value
Data[0]	00：Set successfully 01：Set failed
Data[1]	When set successfully, represents the current baud rate

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	0B	07	17
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	0B	00 07	18

Host sends 1 byte Data: 07, meaning set serial baud rate to 115200 bps

Slave returns 2 bytes Data: 00 07, 00 means set successfully, 07 means current baud rate is 115200 bps.

2） Get Serial Baud Rate Cmd:0C

Gets the device's current serial baud rate. Host sends empty Data field; Slave returns Data field of 1 byte, representing the device's current serial baud rate. The correspondence between the data content and the baud rate is the same as the previous command.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	0C	（Empty）	10
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	0C	07	18

Slave returns 1 byte Data: 07, meaning current serial baud rate is 115200 bps.

3） Restore Factory Settings Cmd:0E

Restores factory settings. The average acquisition count defaults to 1, the serial baud rate defaults to 115200. The Modbus slave address defaults to 1, and the Modbus acquisition integration time defaults to 10. Other spectrometer parameters will not change.

Host sends Data field of 1 byte: 00 means restore all parameters to factory settings, 22 means only restore the serial baud rate to the default value (115200 bps).

Slave returns Data field of 1 byte: 00 means restore successful, 01 means restore failed.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	0E	00	13
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	0E	00	13

Host sends 1 byte Data: 00, meaning restore all parameters to factory settings.

Slave returns 1 byte Data: 00, meaning restore successful.

4） Set TEC Temperature Cmd:12

Sets the TEC temperature (detector temperature). Some devices do not have this function. Host sends Data field of 2 bytes.

Byte	Corresponding Value
Data[0]	00：indicates positive value FF：indicates negative value
Data[1]	Set temperature value (units °C, decimal)

Slave returns Data field of 1 byte: 00 means set successful, 01 means set failed.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	12	FF 0A	21
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	12	00	17

Host sends 2 bytes Data: FF 0A. The first byte (FF) indicates negative value, the second byte (0A) converts to decimal 10. This sets the TEC temperature to -10°C.
Slave returns 1 byte Data: 00, meaning set successful.

5） Get TEC Temperature Cmd:13

Gets the TEC temperature (detector temperature). Some devices do not have this function. Host sends empty Data field; Slave returns Data field of 5 bytes, representing the temperature as ASCII code, formatted like "-10.0". When the content is "XXX.X", it means this device model does not support this temperature detection function.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	13	（Empty）	17
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 09	13	2D 30 34 2E 35	10

Slave returns 5 bytes Data: 2D 30 34 2E 35, which converts to the ASCII string "-04.5", meaning the current detector temperature is -4.5°C.

6） Set Acquisition Average Count Cmd:28

Sets the acquisition average count. After successful setting, whenever the device starts acquisition (whether by software or hardware trigger), it will repeat the acquisition this number of times and return the averaged spectral data. Acquisition time = current integration time × acquisition average count. The average acquisition count range is 1~1024 times.

Host sends Data field of 2 bytes, representing the average acquisition count in decimal.

Slave returns Data field of 1 byte: 00 means set successful, 01 means set failed.

Example:

	Head	Length	Cmd	Data	Checksum
Host Send	AA 55	00 06	28	00 0A	38
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	28	00	2D

Host sends 2 bytes Data: 00 0A, meaning average acquisition count is decimal 10.
Slave returns 1 byte Data: 00, meaning set successful.

7） Get Acquisition Average Count Cmd:2B

Gets the device's current acquisition average count. Host sends empty Data field; Slave returns Data field of 2 bytes, representing the acquisition average count in decimal.

Example:

	Head	Length	Cmd	Data	Checksum
Host Send	AA 55	00 04	2B	（Empty）	2F
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	2B	00 0A	3B

Slave returns 2 bytes Data: 00 0A, which converts to decimal 10, meaning the current acquisition average count is 10.

8） Get Current Integration Time Cmd:41

Gets the device's current integration time. Host sends empty Data field; Slave returns Data representing the current integration time in decimal. The number of bytes and the unit of the integration time are determined by the device attributes.

Example:

	Head	Length	Cmd	Data	Checksum
Host Send	AA 55	00 04	41	（Empty）	45
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	41	00 0A	51

Slave returns 2 bytes Data: 00 0A, which converts to decimal 10, meaning the current integration time is 10 (units depend on device attributes).

9） Get Maximum Integration Time Cmd:42

Gets the device's maximum integration time. Host sends empty Data field; Slave returns Data representing the maximum integration time in decimal. The number of bytes and the unit of the integration time are determined by the device attributes.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	42	（Empty）	46
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	42	FF FF	46

Slave returns 2 bytes Data: FF FF, which converts to decimal 65535, meaning the maximum integration time is 65535.

10） Get Minimum Integration Time Cmd:43

Gets the device's minimum integration time. Host sends empty Data field; Slave returns Data representing the minimum integration time in decimal. The number of bytes and the unit of the integration time are determined by the device attributes.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	43	（Empty）	47
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	43	00 01	4A

Slave returns 2 bytes Data: 00 01, which converts to decimal 1, meaning the minimum integration time is 1.

11） Set GPIO Pin Output Level Cmd:61

Sets the GPIO pin output levels. This function reverts to default upon power loss.

Host sends Data field of 2 bytes. Converted to binary, the LSB represents GPIO0, the second LSB represents GPIO1, and so on. A value of 1 sets the pin high, 0 sets it low.

Slave returns Data field of 1 byte: 00 means set successful, 01 means set failed.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	61	00 03	6A
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	61	00	66

Host sends 2 bytes Data: 00 03, which converts to binary 0011, meaning GPIO0 and GPIO1 are set high, others are low.

2.3.3. Sampling Functions

1） Set Integration Time Cmd:14

For asynchronous acquisition (commands 16 and 23), the integration time must be set first. Synchronous acquisition commands include the integration time and do not require setting it via this command first.

Host sends Data representing the integration time in decimal. The number of bytes and the unit of the integration time are determined by the device attributes.

Slave returns Data field of 1 byte: 00 means set successful, 01 means set failed.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	14	00 32	4C
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	14	00	19

Host sends 2 bytes Data: 00 32, which converts to decimal 50, meaning set integration time to 50. Slave returns 1 byte Data: 00, meaning set successful.

2） Acquire Dark Current (Asynchronous Mode) Cmd:23

Performs CCD asynchronous dark current acquisition. Before starting asynchronous acquisition, the integration time must be set first (command 14). After waiting for the integration time to complete the acquisition, use the Read CCD Acquisition Data command (17) to obtain the acquired data.

Host sends empty Data field; Slave returns Data field of 1 byte: 00 means acquisition successful, 01 means acquisition failed.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	23	（Empty）	27
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	23	00	28

Slave returns 1 byte Data: 00, meaning acquisition successful.

3） Start CCD Scan (Asynchronous Mode) Cmd:16

Starts CCD asynchronous scan. Before starting asynchronous acquisition, the integration time must be set first (command 14). After starting the asynchronous scan, it directly returns success or failure, and the CCD starts acquiring. After the integration time ends, use the Read CCD Acquisition Data command (17) to obtain the acquisition data.

Host sends empty Data field; Slave returns Data field of 1 byte: 00 means acquisition successful, 01 means acquisition failed.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	16	（Empty）	1A
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	16	00	1B

Slave returns 1 byte Data: 00, meaning acquisition successful.

4） Read CCD Acquisition Data Cmd:17

Reads the data acquired by the CCD. Different spectrometers have different CCD pixel counts, so the length of the acquired CCD data varies.

Host sends empty Data field.

Slave returns Data field: Status byte (1 byte) + CCD data (2 * n bytes, n is the pixel length). The status byte 00 means data obtained successfully, FF means data acquisition failed. The CCD data uses 2 bytes per pixel, high byte first, low byte last, corresponding to the intensity data in decimal. Single pixel data range is 0x0000-0xFFFF (decimal 0-65535).

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	17	（Empty）	1B
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	10 05	17	00 2B 2C 2B 67 2B 71 2B B2 .... 3F D8 3F 40	F4

Slave returns 4103 bytes of data. Subtracting the 6 bytes for <Header><Length><Command><Checksum>, the valid data is 4097 bytes. The status byte 00 means data obtained successfully. The remaining 4096 bytes of data correspond to 2048 pixels. The first pixel data is 2B 2C, converting to decimal 11052; the second pixel data is 2B 67, converting to decimal 11111; the second last pixel data is 3F D8, converting to decimal 16344; the last pixel data is 3F 40, converting to decimal 16192.

5） Acquire Dark Current (Synchronous Mode) Cmd:2F

Performs CCD synchronous dark current acquisition and directly returns the acquired data upon completion. Different spectrometers have different CCD pixel counts, so the length of the acquired CCD data varies.

Host sends Data representing the acquisition integration time in decimal. The number of bytes and the unit of the integration time are determined by the device attributes.

Slave returns Data field: Status byte (1 byte) + CCD data (2 * n bytes, n is the pixel length). The status byte 00 means data obtained successfully, FF means data acquisition failed. The CCD data uses 2 bytes per pixel, high byte first, low byte last, corresponding to the intensity data in decimal. Single pixel data range is 0x0000-0xFFFF (decimal 0-65535).

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	2F	00 0A	3F
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	10 05	2F	00 03 A8 03 A6 03 AA 03 A7 ...... 03 9C 03 9E	D7

Host sends 2 bytes Data: 00 0A, converting to decimal 10, meaning acquisition integration time is 10.

The slave returns 2055 bytes of data. Subtracting the 6 bytes for

<Header><Length><Command><Checksum>, the valid data is 2049 bytes. The status byte 00 means data obtained successfully. The remaining 2048 bytes of data correspond to 1024 pixels. The first pixel data is 03 A8, converting to decimal 936; the second pixel data is 03 A6, converting to decimal 934; the second last pixel data is 03 9C, converting to decimal 924; the last pixel data is 03 9E, converting to decimal 926.

6） Start CCD Scan (Synchronous Mode)　　Cmd:1E

Starts CCD synchronous scan and directly returns the acquired data upon completion. Different spectrometers have different CCD pixel counts, so the length of the acquired CCD data varies. When a xenon lamp is connected, using this command for acquisition will turn on the xenon lamp.

Host sends Data representing the acquisition integration time in decimal. The number of bytes and the unit of the integration time are determined by the device attributes.

Slave returns Data field: Status byte (1 byte) + CCD data (2 * n bytes, n is the pixel length). The status byte 00 means data obtained successfully, FF means data acquisition failed. The CCD data uses 2 bytes per pixel, high byte first, low byte last, corresponding to the intensity data in decimal. Single pixel data range is 0x0000-0xFFFF (decimal 0-65535).

Example:

	Head	Length	Cmd	Data	Checksum
Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	1E	00 0A	3F
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	10 05	1E	00 2B 57 2C E1 2E 5F 30 18 31 ...... 12 81 12 49	27

Host sends 2 bytes Data: 00 0A, converting to decimal 10, meaning acquisition integration time is 10.

The slave returns 2055 bytes of data. Subtracting the 6 bytes for <Header><Length><Command><Checksum>, the valid data is 2049 bytes. The status byte 00 means data obtained successfully. The remaining 2048 bytes of data correspond to 1024 pixels. The first pixel data is 2B 57, converting to decimal 11095; the second pixel data is 2C E1, converting to decimal 11489; the second last pixel data is 12 81, converting to decimal 4737; the last pixel data is 12 49, converting to decimal 4681.

7） Enable External Trigger Acquisition Function　　Cmd:1F

After enabling, when the external trigger pin generates a valid level change, it acquires spectral data and directly returns the CCD data after acquisition is complete. The external trigger function enables the xenon lamp by default, meaning when a xenon lamp is connected, it will turn on during external trigger acquisition.

Host sends Data representing the acquisition integration time in decimal. The number of bytes and the unit of the integration time are determined by the device attributes.

Slave returns Data field of 1 byte: 00 means enable successful, 01 means enable failed.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	1F	00 0A	2F
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	1F	00	24

Host sends 2 bytes Data: 00 0A, converting to decimal 10, meaning acquisition integration time is 10.

Slave returns 1 byte Data: 00, meaning enable successful.

8） Disable External Trigger Acquisition Function Cmd:F1

Disables the external trigger acquisition function.

Host sends empty Data field; Slave returns Data field of 1 byte: 00 means disable successful, 01 means disable failed.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	F1	（Empty）	F5
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	F1	00	F6

Slave returns 1 byte Data: 00, meaning disable successful.

2.3.4. Calibration Coefficient Functions

1） Get Wavelength Calibration Coefficients Cmd:55

Wavelength calibration coefficients are required to convert spectrometer pixels to wavelength.

Host sends Data field fixed as 00 01; Slave returns Data field of 514 bytes. The first two bytes are the packet number, bytes 17 to 32 are the wavelength calibration coefficients. Every 4 bytes form one calibration coefficient, [Low byte first, high byte last]{.underline}, convert hexadecimal to single-precision float.

The relationship between pixel and wavelength is as follows:

$$y = ax^3 + bx^2 + cx + d$$

y： Wavelength

x： Pixel, starting from 0

Coefficient a is bytes 17 to 20；

Coefficient b is bytes 21 to 24；

Coefficient c is bytes 25 to 28；

Coefficient d is bytes 29 to 32。

Example:

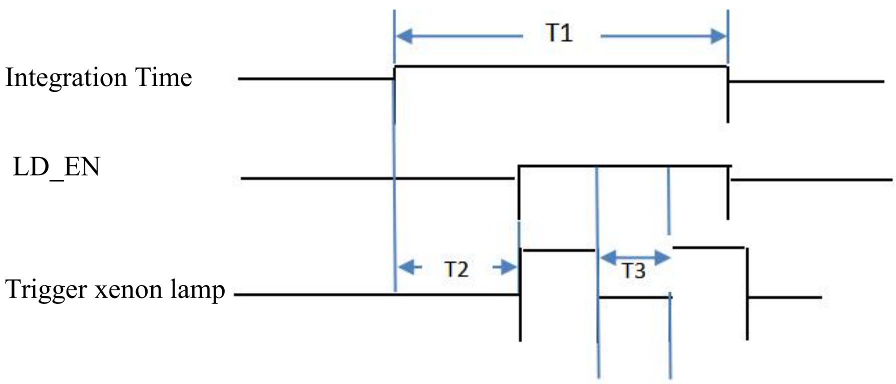
Host Send	Head	Length	Cmd	Data	Checksum

	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	55	00 01	5C
Slave Return	AA 55	02 06	55	00 01 00 00 00 86 07 F6 14 00 00 00 04 00 00 00 11 E5 F2 AF 30 AF 35 37 74 3A D9 3E 1B C7 42 43 AD BE 44 64 2C 27 93	1D

In the slave return data, bytes 17-20 are 11 E5 F2 AF (corresponding to coefficient a), which converts to the single-precision float -4.4182305036777336e-10.
Note: Low byte first, high byte last means little-endian order, pay attention during conversion.

2.3.5. Xenon Lamp Functions

Pin marked as "Lamp/LD Enable pin" in the Electrical Interface section is the current spectrometer's xenon lamp trigger pin. When the spectrometer receives command 1E or an external trigger, it can pull the xenon lamp pin high. In these cases, while the spectrometer starts integration, the xenon lamp pin waits for the xenon lamp delay trigger time and then pulls high (the xenon lamp delay trigger time can be 0, controlled by command 24). The number of times the xenon lamp pin is pulled high and the interval between high levels are controlled by the Set Xenon Lamp Flash Count (F2) and Set Xenon Lamp Frequency (F3) commands. After the integration time ends, the xenon lamp control pin is pulled low.



Note:

1. After enabling the xenon lamp, ensure the xenon lamp triggers within the integration time.
2. T1 is the set integration time, T2 is the set xenon lamp delay trigger time (i.e., delay time before triggering the synchronous xenon signal). T3 is 1 / (xenon lamp frequency).
3. Xenon lamp delay trigger time + (Xenon lamp trigger count - 1) * (1000 / Xenon lamp trigger frequency) <= Integration time.

1） Set Xenon Lamp Delay Trigger Time　　Cmd:24

After the spectrometer starts acquisition, after the xenon lamp delay trigger time, pull the xenon lamp trigger pin high. Setting is retained after power loss.

Host sends Data field of 2 bytes, representing the delay time to trigger the xenon flash after starting acquisition, in decimal, units μs.

Slave returns Data field of 1 byte: 00 means set successful, 01 means set failed, FF means function not available.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	24	00 00	2A
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	24	00	29

Host sends 2 bytes Data: 00 00, converting to decimal 0, meaning set delay trigger time to 0, i.e., trigger xenon lamp synchronously with acquisition.
Slave returns 1 byte Data: 00, meaning set successful.

2） Set Xenon Lamp Flash Count Cmd:F2

Sets the xenon lamp flash count. This is the number of times the xenon lamp pin is pulled high during the integration time. Setting is retained after power loss.

Host sends Data field of 2 bytes, representing the set xenon lamp flash count, in decimal, value range 0x0000-0xFFFF (decimal 0-65535).

Slave returns Data field of 1 byte: 00 means set successful, 01 means set failed, FF means function not available.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	F2	00 01	F9
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	F2	00	F7

Host sends 2 bytes Data: 00 01, converting to decimal 1, meaning set xenon lamp flash count to 1.
Slave returns 1 byte Data: 00, meaning set successful.

3） Set Xenon Lamp Frequency Cmd:F3

Sets the xenon lamp trigger frequency, unit Hz. This parameter affects the proportion of low level within one cycle. For example, setting the xenon lamp frequency to 100 means the interval between xenon lamp high levels during the integration time is (1000/100 = 10ms). Setting is retained after power loss.

Host sends Data field of 2 bytes, representing the set xenon lamp frequency, in decimal, value range 00 00-01 F4 (decimal 0-500).

Slave returns Data field of 1 byte: 00 means set successful, 01 means set failed, FF means function not available.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 06	F3	00 0A	03
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 05	F3	00	F8

Host sends 2 bytes Data: 00 0A, converting to decimal 10, meaning set xenon lamp frequency to 10Hz.
Slave returns 1 byte Data: 00, meaning set successful.

4） Get Xenon Lamp Attributes Cmd:F5

Gets the xenon lamp attributes, including flash count, frequency, and delay trigger time.

Host sends empty Data field; Slave returns Data field of 6 bytes. Bytes 1-2 represent the xenon lamp flash count, bytes 3-4 represent the xenon lamp frequency, bytes 5-6 represent the xenon lamp delay trigger time. If the valid data is all FF FF, it means this function is not available.

Example:

Host Send	Head	Length	Cmd	Data	Checksum
	AA 55	00 04	F5	（Empty）	F9
Slave Return	Head	Length	Cmd	Data	Checksum
	AA 55	00 0A	F5	00 01 00 0A 00 00	0A

Slave returns 6 bytes Data: Bytes 1-2 are 00 01, meaning xenon lamp flash count is 1; bytes 3-4 are 00 0A, meaning xenon lamp frequency is 10Hz; bytes 5-6 are 00 00, meaning xenon lamp delay trigger time is 0μs.

3.ATP Series Optical Spectrometer Modbus Communication Protocol

3.1.Modbus Protocol and Command Format

The following protocol content is based on the national standard "Industrial Automation Network Specification Based on Modbus Protocol - Part 1: Modbus Application Protocol" (GB/T 19582.1-2008).

1）Modbus Protocol Supported Function Codes

0x01: Read Multiple Coils;
0x02: Read Multiple Discrete Inputs;
0x03: Read Multiple Holding Registers;
0x04: Read Multiple Input Registers;
0x05: Write Single Coil;
0x06: Write Single Holding Register;
0x0F: Write Multiple Coils;
0x10: Write Multiple Holding Registers.

3.2.Register Preview

Table 4.1 Register Information

NO.	Register Name	Register Type	Register Address	Function Description	Note
1	Collect Dark Background Data	Coil	0x0000	Set to ON to collect dark background data. Automatically resets after collection.	Dark background data collection takes priority. If both are enabled simultaneously, only dark background data collection will be executed.
2	Collect Spectral Data	Coil	0x0001	Set to ON to collect spectral data. Automatically resets after collection.	
3	Subtract Dark Background Data	Coil	0x0002	Set to ON: Measurement point data equals spectral data minus dark background data for the corresponding pixel. Set to OFF: Measurement point data equals spectral data for the corresponding pixel.	

4	Minimum Effective Wavelength	Input Register	0x0000	Query the spectral instrument's wavelength range.	
	Maximum Effective Wavelength	Input Register	0x0001		
5	Device Pixel Length	Input Register	0x0002	Obtain the device pixel length.	
6	Pixel Corresponding to Measurement Point 1	Input Register	0x0020	Automatically matches the corresponding pixel point of the spectral instrument based on the wavelengths set for measurement points 1 to 10. If the set wavelength is less than the minimum effective wavelength, the corresponding pixel point is 0; if the set wavelength is greater than the maximum effective wavelength, the corresponding pixel point is 511.	
	Pixel Corresponding to Measurement Point 2	Input Register	0x0021		
	……	Input Register	……		
	Pixel Corresponding to Measurement Point 10	Input Register	0x0029		
7	Data of Measurement Point 1	Input Register	0x0090	Data source determined by the "Subtract Dark Background Data" setting: If set to ON, data is spectral data minus dark background data. If set to OFF, data is spectral data.	
	Data of Measurement Point 2	Input Register	0x0091		
	……	Input Register	……		
	Data of Measurement Point 10	Input Register	0x0099		
8	Spectral Data of Pixel 0	Input Register	0x1000	Data collected from all CCD pixels during the last spectral data collection operation. If the number of pixels varies by model, the address of the last pixel will differ. Note: Each	
	Spectral Data of Pixel 1	Input Register	0x1001		
	……	Input Register	……		

				pixel occupies one register, and one register contains two bytes.	
9	Dark Background Data of Pixel 0	Input Register	0x4000	Data collected from all CCD pixels during the last dark background data collection operation. If the number of pixels varies by model, the address of the last pixel will differ. Note: Each pixel occupies one register, and one register contains two bytes.	
	Dark Background Data of Pixel 1	Input Register	0x4001		
	……	Input Register	……		
10	Slave Address	Holding Register	0x0000	Device slave address. Stored after power-off.	
11	Baud Rate	Holding Register	0x0001	Device baud rate.	
12	Integration Time	Holding Register	0x0002	Device integration time. Stored after power-off.	
		Holding Register	0x0003		
13	Averaging Count	Holding Register	0x0004	Device averaging count. Stored after power-off.	
14	Wavelength of Measurement Point 1	Holding Register	0x0020	Device measurement point wavelengths, up to 10. Stored after power-off.	
	Wavelength of Measurement Point 2	Holding Register	0x0021		
	……	Holding Register	……		
	Wavelength of Measurement Point 10	Holding Register	0x0029		
15	Xenon Lamp Trigger Count	Holding Register	0x0040		
16	Xenon Lamp Trigger Frequency	Holding Register	0x0041		
17	Xenon Lamp Delay Trigger Time	Holding Register	0x0042		

Note：

Dark background data collection takes priority. If both are enabled simultaneously, only dark background data collection will be executed.

3.3.Modbus Command Application Examples

The protocol format and CRC check values in the following tables must be calculated based on the actual commands.

In the examples provided in this chapter, the address field defaults to 01. If the user has set a slave address, the address field in this chapter must be modified to the user-set slave address. The slave address is retained after power-off and only needs to be set once. If necessary, it can be set again. Users are advised to memorize the set slave address to avoid forgetting it later!

In the command examples in this chapter, the content highlighted in red represents variable data that may change based on the actual conditions of the spectrometer!

3.3.1.Read Multiple Coils

1）Get Current Collection Status

Response register values: 00 indicates idle state, 01 indicates collecting dark background data, 02 indicates collecting spectral data.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	01	00 00	00 03	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	01	01	00	/

3.3.2.Read Multiple Holding Registers

As this section involves reading Holding Registers, the examples provided use the Starting Address as the parameter storage address, and the Number of Registers indicates the quantity of parameters that can be stored.

1）Get Slave Address

Response Register Value (2 bytes): 00 01 represents the device slave address. The device slave address range is 1 to 247, and the factory default slave address is 01.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	03	00 00	00 01	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	03	02	00 01	/

2）Get Current Device Serial Port Baud Rate

Response Register Value (2 bytes): 00 07 indicates the serial port baud rate is 115200.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	03	00 01	00 01	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	03	02	00 07	/

Obtain the baud rate of the communication serial port. The change takes effect after power-off and restart. The sent command data is 2 bytes, ranging from 0x0001 to 0x0010, corresponding to the following baud rates:
1 (0x0001): 9600 baud
2 (0x0002): 14400 baud
3 (0x0003): 19200 baud
4 (0x0004): 38400 baud
5 (0x0005): 56000 baud
6 (0x0006): 57600 baud
7 (0x0007): 115200 baud (default)
8 (0x0008): 128000 baud
9 (0x0009): 230400 baud
10 (0x000A): 256000 baud
11 (0x000B): 460800 baud
12 (0x000C): 500000 baud
13 (0x000D): 512000 baud
14 (0x000E): 600000 baud
15 (0x000F): 750000 baud
16 (0x0010): 921600 baud

3）Get Integration Time

Response Register Value (4 bytes): 00 00 00 0A indicates an integration time of 10ms. The factory default integration time is 10ms (i.e., 00 00 00 0A). The integration time is stored after power-off.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	03	00 02	00 02	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	03	04	00 00 00 0A	/

4）Get Averaging Count

Response Register Value (2 bytes): 00 01 indicates an averaging count of 1 time. The averaging count ranges from 1 to 1024 times. The factory default averaging count is 1 time.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	03	00 04	00 01	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	03	02	00 01	/

5）Get Measurement Point Wavelength

The Starting Address in the request ranges from 0x0020 to 0x0029, and the Number of Registers ranges from 0x0001 to 0x000A, indicating that the wavelengths of 10 measurement points can be obtained at once, or individual or multiple measurement point wavelengths can be retrieved.

Example:

- Get all measurement point wavelengths:

Request Example: 01 03 00 20 00 0A

- Get the first measurement point wavelength:

Request Example: 01 03 00 20 00 01

- Get the third measurement point wavelength:

Request Example: 01 03 00 22 00 01

And so on. Data can be retrieved according to actual needs.

Response Register Value (2 to 20 bytes): The specific Byte Count of the response Register Value varies based on the Number of Registers set in the request command.
For example, a response of 00 C1 (i.e., the Register Value in hexadecimal is 0x00C1, which converts to 193 in decimal) indicates that the wavelength of the first measurement point is 193 nm. Similarly, 00 C4 indicates that the wavelength of the second measurement point is 196 nm. The wavelengths of the remaining measurement points follow the same pattern.

Address Field	Function Code	Starting Address	Number of Registers	CRC
01	03	00 20	00 0A	/
Address Field	Function Code	Byte Count	Register Value	CRC
01	03	14	00 C1 00 C4 01 20……	/

6）Get Xenon Lamp Maximum Trigger Count

Response Register Value (2 bytes): 00 0A indicates the xenon lamp can be triggered up to 10 times during the integration time. The factory default trigger count is 1. Some spectrometer models do not have xenon lamp triggering functionality, so this command is not available for them.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	03	00 40	00 01	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	03	02	00 0A	/

7）Get Xenon Lamp Trigger Frequency

Response Register Value (2 bytes): 00 0A indicates the xenon lamp is triggered 10 times per second. The factory default trigger frequency is 1. Some spectrometer models do not have xenon lamp triggering functionality, so this command is not available for them.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	03	00 41	00 01	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	03	02	00 0A	/

8）Get Xenon Lamp Delay Trigger Time

Response Register Value (2 bytes): 00 0A indicates a xenon lamp delay trigger time of 10 µs. The factory default delay trigger time is 0 µs. Some spectrometer models do not have xenon lamp triggering functionality, so this command is not available for them.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	03	00 42	00 01	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	03	02	00 0A	/

9）Get Xenon Lamp Properties

This command is a combination of the three commands: "Get Xenon Lamp Maximum Trigger Count," "Get Xenon Lamp Trigger Frequency," and "Get Xenon Lamp Delay Trigger Time." Response Register Value (6 bytes): The first 00 0A represents the maximum trigger count; the second 00 0A represents a trigger frequency of 10; the third 00 0A represents a delay trigger time of 10 µs. Some spectrometer models do not have xenon lamp triggering functionality, so this command is not available for them.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	03	00 40	00 03	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	03	06	00 0A 00 0A 00 0A	/

3.3.3.Read Multiple Input Registers

1）Get Device Effective Wavelength Range

Response Register Value (4 bytes), varies based on the spectrometer's actual wavelength range: 00 C1 indicates a minimum wavelength of 193 nm; 04 1F indicates a maximum wavelength of 1055 nm. Thus, the spectrometer's wavelength range is 193-1055 nm.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	04	00 00	00 02	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	04	04	00 C1 04 1F	/

2）Get Device Pixel Length

Response Register Value (2 bytes) varies based on the spectrometer's actual pixel length: 08 00 indicates a device pixel length of 2048. Different spectrometers have different pixel counts.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	04	00 02	00 01	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	04	02	08 00	/

3）Get Pixel Corresponding to Measurement Point

The Starting Address in the request ranges from 0x0020 to 0x0029, and the Number of Registers ranges from 0x0001 to 0x000A, indicating that the pixels corresponding to the wavelengths of 10 measurement points can be obtained at once, or individual or multiple pixels corresponding to measurement point wavelengths can be retrieved.

Example:

- Get pixels corresponding to all measurement point wavelengths:

Request Example: 01 04 00 20 00 0A

- Get the pixel corresponding to the first measurement point wavelength:

Request Example: 01 04 00 20 00 01

- Get the pixel corresponding to the third measurement point wavelength:

Request Example: 01 04 00 22 00 01

And so on. Data can be retrieved according to actual needs.

Response Register Value (2 to 20 bytes): The Number of Registers in the Request Example varies based on the desired data, which affects the Byte Count and Register Value in the Response Example.

For example, a response of 00 C1 indicates that the pixel corresponding to the first measurement point wavelength is 00 C1; 04 1F indicates that the pixel corresponding to the second measurement point wavelength is 04 1F. Subsequent measurement points follow the same pattern.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	04	00 20	00 0A	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	04	14	00 C1 04 1F……	/

4）Get Measurement Point Data

The Starting Address in the request ranges from 0x0090 to 0x0099, and the Number of Registers ranges from 0x0001 to 0x000A, indicating that the spectral intensity corresponding to 10 measurement points can be obtained at once, or individual or multiple spectral intensities corresponding to measurement points can be retrieved.

Example:

- Get spectral intensities for all measurement points:

Request Example: 01 04 00 90 00 0A

- Get spectral intensity for the first measurement point:

Request Example: 01 04 00 90 00 01

- Get spectral intensity for the third measurement point:

Request Example: 01 04 00 92 00 01
And so on. Data can be retrieved according to actual needs.

Response Register Value (2 to 20 bytes): The Number of Registers in the Request Example varies based on the desired data, which affects the Byte Count and Register Value in the Response Example.
For example, a response of 01 22 indicates that the data intensity of the first measurement point is 290; 01 25 indicates that the data intensity of the second measurement point is 293. Subsequent measurement points follow the same pattern.
If "Subtract Dark Background Data" is set to OFF, the retrieved data is the collected spectral data.
If "Subtract Dark Background Data" is set to ON, the retrieved data is the collected spectral data minus the dark background data.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	04	00 90	00 0A	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	04	14	01 22 01 25……	/

5）Get Pixel Spectral Data

Retrieve data collected from all pixels. Since a maximum of 125 register values can be obtained at once, and the number of pixels varies by spectrometer model, the last pixel's register address may differ. Therefore, multiple requests are needed, with consecutive Starting Addresses. All returned data must be concatenated to form the complete spectral data collected by the spectrometer.
The Starting Address in the request ranges from 0x1000 to 0x00xx (varies by device), and the Number of Registers ranges from 0x0001 to 0x007D, indicating that the intensity values for up to 125 pixels can be retrieved at once, or individual/multiple pixel intensities can be obtained.

Example:
Get intensities for all pixels:
Request Example: 01 04 10 00 00 7D
Get intensity for the first pixel:
Request Example: 01 04 10 00 00 01
Get intensity for the third pixel:
Request Example: 01 04 10 02 00 01
And so on. Data can be retrieved according to actual needs.

Response Register Value (2 to 250 bytes): The Number of Registers in the Request Example varies based on the desired data, which affects the Byte Count and Register Value in the Response Example.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	04	10 00	00 7D	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	04	FA	XX XX XX XX……	/

6）Get Pixel Dark Current Data

Retrieve dark current data collected from all pixels. Since a maximum of 125 register values can be obtained at once, and the number of pixels varies by spectrometer model, the last pixel's register address may differ. Therefore, multiple requests are needed, with consecutive Starting Addresses. All returned data must be concatenated to form the complete dark current data collected by the spectrometer.

The Starting Address in the request ranges from 0x1000 to 0x00xx (varies by device), and the Number of Registers ranges from 0x0001 to 0x007D, indicating that the intensity values for up to 125 pixels can be retrieved at once, or individual/multiple pixel intensities can be obtained.

Example:

- Get intensities for all pixels:

Request Example: 01 04 40 00 00 7D

- Get intensity for the first pixel:

Request Example: 01 04 40 00 00 01

- Get intensity for the third pixel:

Request Example: 01 04 40 02 00 01

And so on. Data can be retrieved according to actual needs.

Response Register Value (2 to 250 bytes): The Number of Registers in the Request Example varies based on the desired data, which affects the Byte Count and Register Value in the Response Example.

Request Example	Address Field	Function Code	Starting Address	Number of Registers	CRC
	01	04	40 00	00 7D	/
Response Example	Address Field	Function Code	Byte Count	Register Value	CRC
	01	04	FA	XX XX XX XX……	/

3.3.4.Write Single Coil

1）Set Collect Dark Background Data

Request/Response Register Value (2 bytes): Automatically resets to OFF (Register Value 00 00) after collection is completed.

Request Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	05	00 00	FF 00	/
Response Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	05	00 00	FF 00	/

2）Set Collect Spectral Data

Request/Response Register Value (2 bytes): Automatically resets to OFF (Register Value 00 00) after collection is completed.

Request Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	05	00 01	FF 00	/
Response Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	05	00 01	FF 00	/

3）Set Subtract Dark Background Data

Request/Response Register Value (2 bytes): Register Value set to FF 00 indicates ON, meaning measurement point data is derived from spectral data minus dark background data.

Request Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	05	00 02	FF 00	/
Response Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	05	00 02	FF 00	/

3.3.5.Write Single Holding Register

1）Set Device Slave Address

Request/Response Register Value (2 bytes): Register Value 00 01 indicates the set device slave address is 00 01. The device slave address range is 1 to 247, and the factory default slave address is 1.

Request Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 00	00 01	/
Response Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 00	00 01	/

2）Set Serial Port Baud Rate

Request/Response Register Value (2 bytes): Register Value 00 07 indicates the set serial port baud rate is 115200. The factory default serial port baud rate is 115200.

Request Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 01	00 07	/
Response Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 01	00 07	/

The baud rate of the communication serial port takes effect after power-off and restart. The sent command data is 2 bytes, ranging from 0x0001 to 0x0010, corresponding to the following baud rates:

1 (0x0001): 9600 baud
2 (0x0002): 14400 baud
3 (0x0003): 19200 baud
4 (0x0004): 38400 baud
5 (0x0005): 56000 baud
6 (0x0006): 57600 baud
7 (0x0007): 115200 baud (default)
8 (0x0008): 128000 baud
9 (0x0009): 230400 baud
10 (0x000A): 256000 baud
11 (0x000B): 460800 baud
12 (0x000C): 500000 baud
13 (0x000D): 512000 baud
14 (0x000E): 600000 baud
15 (0x000F): 750000 baud
16 (0x0010): 921600 baud

3）Set Averaging Count

Request/Response Register Value (2 bytes): Register Value 00 05 indicates the set averaging count is 5 times. The averaging count range is 1 to 1024 times, and the factory default averaging count is 1 time.

Request Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 04	00 05	/
Response Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 04	00 05	/

4）Set Xenon Lamp Trigger Count

Request/Response Register Value (2 bytes): Register Value 00 0A indicates the set maximum trigger count for the xenon lamp during integration time is 10 times. The factory default trigger count is 1 time. Some spectrometer models do not have xenon lamp triggering functionality, so this command is not available for them.

Request Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 40	00 0A	/
Response Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	0040	00 0A	/

5）Set Xenon Lamp Trigger Frequency

Request/Response Register Value (2 bytes): Register Value 00 0A indicates the set xenon lamp trigger frequency is 10. The factory default trigger frequency is 1. Some spectrometer models do not have xenon lamp triggering functionality, so this command is not available for them.

Request Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 41	00 0A	/
Response Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 41	00 0A	/

6）Set Xenon Lamp Delay Trigger Time

Request/Response Register Value (2 bytes): Register Value 00 0A indicates the set xenon lamp delay trigger time is 10 µs. The factory default delay trigger time is 0 µs. Some spectrometer models do not have xenon lamp triggering functionality, so this command is not available for them.

Request Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 42	00 0A	/
Response Example	Address Field	Function Code	Register Address	Register Value	CRC
	01	06	00 42	00 0A	/

3.3.6.Write Multiple Coils

1）Collect Spectral Data and Subtract Dark Background Data for Measurement Points

After sending the command, the spectrometer first collects dark background data, then collects spectral data, and finally subtracts the dark background data from the spectral data.

Request Example	Address Field	Function Code	Register Starting Address	Number of Registers	Byte Count	Register Value	CRC
	01	0F	00 00	00 03	01	07	/
Response Example	Address Field	Function Code	Register Starting Address		Number of Registers		CRC
	01	0F	00 00		00 03		/

3.3.7.Write Multiple Holding Registers

1）Set Measurement Point Wavelengths

Set Measurement Point Wavelengths
The Starting Address in the request ranges from 0x0020 to 0x0029, and the Number of Registers ranges from 0x0001 to 0x000A, indicating that the wavelengths of 10 measurement points can be set at once, or individual or multiple measurement point wavelengths can be set.

Example:

- Set all measurement point wavelengths:

Request Example: 01 10 00 20 00 0A 14 00 0A 00 64 00 C8 01 2C 01 90 01 F4 02 58 02 BC 03 20 03 84

- Set 2 measurement point wavelengths:

Request Example: 01 10 00 20 00 02 04 00 64 00 C8
And so on. Data can be set according to actual needs.
Register Value in Request Example: 00 C1 indicates setting the first measurement point wavelength to 193 nm; 00 D1 indicates setting the second measurement point wavelength to 209 nm. This command can also set different numbers of measurement point wavelengths by modifying the Number of Registers, Byte Count, and Register Value in the Request Example.

Request Example	Address Field	Function Code	Register Starting Address	Number of Registers	Byte Count	Register Value	CRC
	01	10	00 20	00 0A	14	00 C1 00 D1...	/
Response Example	Address Field	Function Code	Register Starting Address		Number of Registers		CRC
	01	10	00 20		00 0A		/

2）Set Integration Time

Register Value 00 00 00 0A in the Request Example indicates setting the integration time to 10 ms. The factory default integration time is 10 ms, and the integration time is stored after power-off.

Request Example	Address Field	Function Code	Register Starting Address	Number of Registers	Byte Count	Register Value	CRC
	01	10	00 02	00 02	04	00 00 00 0A	/
Response Example	Address Field	Function Code	Register Starting Address		Number of Registers		CRC
	01	10	00 02		00 02		/

3.4.Modbus Spectral Data Acquisition Process

3.4.1.Command Acquisition Process

After system power-on, reading the intensity of specified measurement points should follow the example steps below:

- Set device communication parameters (including address code; these parameters are non-volatile and remain effective after the first setting)

Set address code to 01: 01 06 00 00 00 01 48 0A

- Set scanning parameters (including integration time, averaging count, measurement point wavelengths; these parameters are non-volatile)

(1) Set integration time to 10 ms: 01 10 00 02 00 02 04 00 00 00 0A F2 71

(2) Set averaging count to 5 times: 01 06 00 04 00 05 08 08

(3) Set all measurement point wavelengths (up to 10 measurement points can be set): 01 10 00 20 00 0A 14 00 0A 00 64 00 C8 01 2C 01 90 01 F4 02 58 02 BC 03 20 03 84

- Set whether to subtract dark background data (select one of the following commands; this setting is volatile)

Enable dark background subtraction: 01 05 00 02 FF 00

Disable dark background subtraction: 01 05 00 02 00 00

- Collect dark current

Set spectrometer collection state, initiate dark current collection: 01 05 00 00 FF 00 8C 3A

- Collect spectral data

Set spectrometer collection state, trigger spectral collection: 01 05 00 01 FF 00 DD FA

- Obtain measurement point intensities and save to host

Get all measurement point intensities: 01 04 00 90 00 0A

3.4.2.Quick Acquisition Process

After system power-on, reading the intensity of specified measurement points should follow the example steps below:

Step 1: Set device communication parameters (including address code; these parameters are non-volatile and remain effective after the first setting)
Set address code to 01: 01 06 00 00 00 01 48 0A

Step 2: Set scanning parameters (including integration time, averaging count, measurement point wavelengths; these parameters are non-volatile)
(1) Set integration time to 10 ms: 01 10 00 02 00 02 04 00 00 00 0A F2 71
(2) Set averaging count to 5 times: 01 06 00 04 00 05 08 08
(3) Set all measurement point wavelengths (up to 10 measurement points can be set): 01 10 00 20 00 0A 14 00 0A 00 64 00 C8 01 2C 01 90 01 F4 02 58 02 BC 03 20 03 84

Step 3: Quick acquisition
Set spectrometer quick acquisition (collect spectral data and subtract dark background data for measurement points): 01 0F 00 00 00 03 01 07

Step 4: Obtain measurement point intensities and save to host
Get all measurement point intensities: 01 04 00 90 00 0A

4. ATP Series Fiber Optic Spectrometer Electrical Interface

The ATP2000P and ATP2002 models feature RS232-level serial data receive/transmit pins, which require customization.

The Ext_trigger_in pin serves as an external trigger input, designed to work with the external triggering function.

The Lamp/LD Enable pin is a light source control pin, used to trigger and manage the light source.

4.1.ATP1010

The module utilizes a dual-row 10-pin box header and a USB Type-C interface for system power supply and communication. The detailed pin descriptions are as follows:

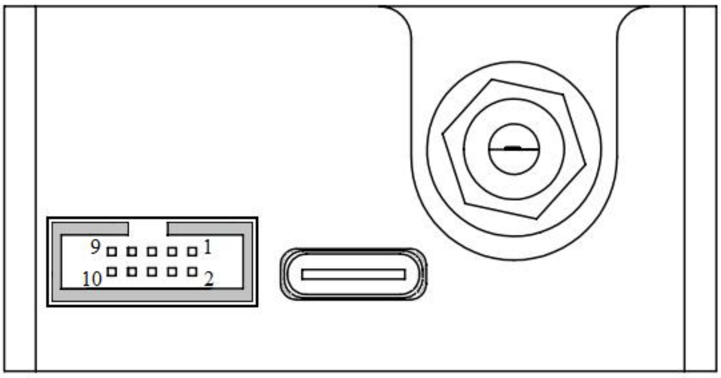


Table 5.1 Electrical Pin Definitions

Pin	Pin Definitions	I/O	Description
1	GND	/	Ground
2	VCC	/	Power Supply Voltage, 5V ±0.5V
3	SPI_MOSI	Output	SPI Master Out Slave In (MOSI) signal for communication with SPI peripherals
4	SPI_SCK	Output	SPI Clock signal for communication with SPI peripherals
5	SPI_MISO	Input	SPI Master In Slave Out (MISO) signal for communication with SPI peripherals
6	SPI1_NSS	Output	SPI Chip/Device Select signal for communication with SPI peripherals
7	Lamp/LD Enable	Output	Light source (Xenon lamp/Laser) control pin output, 5V TTL level
8	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
9	TTL_TX	Output	Serial data transmission pin, TTL level, 3.3V TTL level
10	TTL_RX	Input	Serial data reception pin, TTL level, 3.3V TTL level

4.2.ATP1030

The module utilizes a dual-row 10-pin fenced header and a USB Type-C interface for system power supply and communication. The specific pin descriptions are as follows:

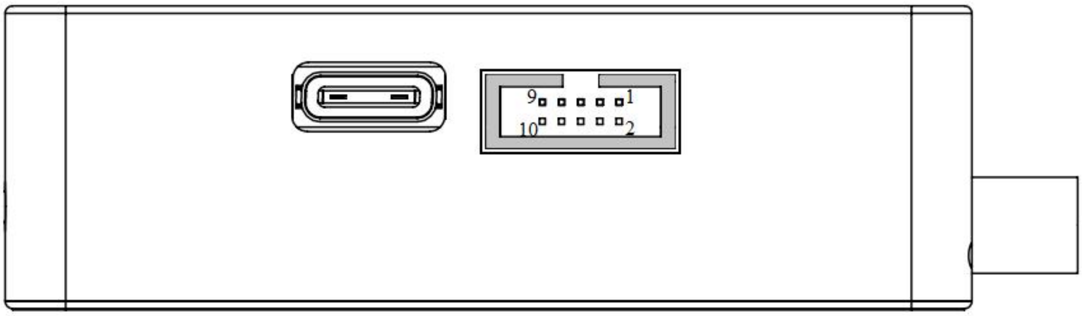


Table 5.2 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	GND	/	Ground
2	VCC	/	Power Supply Voltage, 5V±0.5V
3	SPI_MOSI	Output	SPI Master Output Slave Input (MOSI) signal for communication with SPI peripheral devices
4	SPI_SCK	Output	SPI clock signal for communication with SPI peripheral devices
5	SPI_MISO	Input	SPI Master Input Slave Output (MISO) signal for communication with SPI peripheral devices
6	SPI1_NSS	Output	SPI chip/device select signal for communication with SPI peripheral devices
7	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin output, 5V TTL level
8	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
9	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level
10	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level

4.3.ATP2002/ATP303X

The module utilizes a dual-row 30-pin fenced header and a USB Type-C interface for system power supply and communication. The specific pin descriptions are as follows:

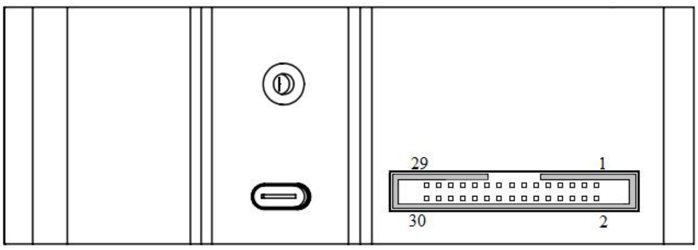


Table5.3 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level
2	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level

3	VCC	/	Power, 5V±0.5V
4	GPIO2	Output	General-purpose software-programmable digital output port, 3.3V TTL level
5	NC	/	No connection
6	GND	/	Ground
7	NC	/	No connection
8	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
9	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
10	GPIO1	Output	General-purpose software-programmable digital output port, 3.3V TTL level
11	NC	/	No connection
12	GPIO3	Output	General-purpose software-programmable digital output port, 3.3V TTL level
13	NC	/	No connection
14	GPIO10	Output	General-purpose software-programmable digital output port, 3.3V TTL level
15	GPIO4	Output	General-purpose software-programmable digital output port, 3.3V TTL level
16	GPIO11	Output	General-purpose software-programmable digital output port, 3.3V TTL level
17	GPIO5	Output	General-purpose software-programmable digital output port, 3.3V TTL level
18	NC	/	No connection
19	NC		No connection
20	GND	/	Ground
21	GPIO6	Output	General-purpose software-programmable digital output port, 3.3V TTL level
22	GND	/	Ground
23	Analog Out (0-5V)	Output	12-bit programmable analog output, 0-5V voltage range
24	NC	/	No connection
25	GPIO7	Output	General-purpose software-programmable digital output port, 3.3V TTL level
26	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin, output, 5V TTL level
27	GPIO8	Output	General-purpose software-programmable digital output port, 3.3V TTL level
28	GND	/	Ground
29	GPIO9	Output	General-purpose software-programmable digital output port, 3.3V TTL level
30	GND	/	Ground

4.4.ATP2000P/H

The module utilizes a dual-row 20-pin fenced header and a USB 2.0 Type-B interface for system power supply and communication. The specific pin descriptions are as follows:

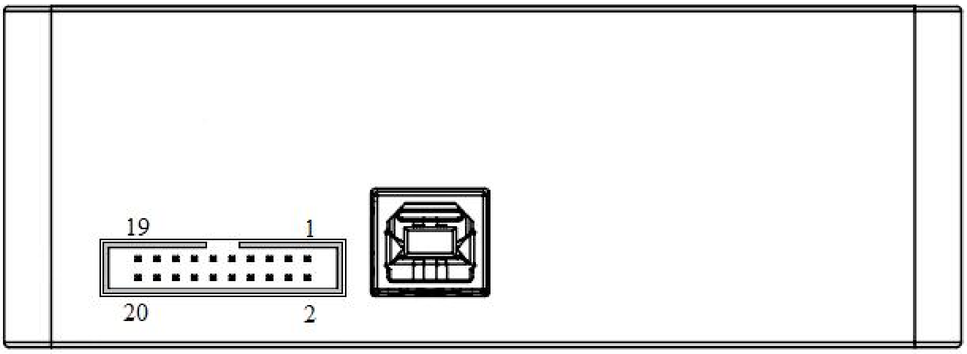


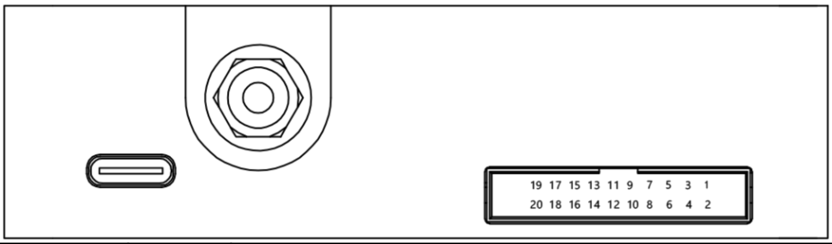
Table5.4 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	GND	/	Ground
2	VCC	/	Power, 5V±0.5V
3	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level
4	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level
5	NC	/	No connection
6	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin, output, 5V TTL level
7	NC	/	No connection
8	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
9	SPI_MOSI	Output	SPI Master Output Slave Input (MOSI) signal for communication with SPI peripheral devices
10	SPI_SCK	Output	SPI clock signal for communication with SPI peripheral devices
11	SPI_CS	Output	SPI Master Input Slave Output (MISO) signal for communication with SPI peripheral devices
12	SPI_MISO	Input	SPI chip/device select signal for communication with SPI peripheral devices
13	GPIO1	Output	General-purpose software-programmable digital output port, 3.3V TTL level
14	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
15	GPIO3	Output	General-purpose software-programmable digital output port, 3.3V TTL level
16	GPIO2	Output	General-purpose software-programmable digital output port, 3.3V TTL level
17	GPIO5	Output	General-purpose software-programmable digital output port, 3.3V TTL level
18	GPIO4	Output	General-purpose software-programmable digital output port, 3.3V TTL level

19	GPIO7	Output	General-purpose software-programmable digital output port, 3.3V TTL level
20	GPIO6	Output	General-purpose software-programmable digital output port, 3.3V TTL level

4.5.ATP2400

The module utilizes a dual-row 20-pin fenced header and a USB Type-C interface for system power supply and communication. The specific pin descriptions are as follows:Table5.5　Electrical Pins



Pin	Pin Definitions	I/O	Description
1	VCC	/	Power, 5V±0.5V
2	GND	/	Ground
3	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level
4	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level
5	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin, output, 5V TTL level
6	NC	/	No connection
7	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
8	NC	/	No connection
9	SPI_SCK	Output	SPI clock signal for communication with SPI peripheral devices
10	SPI_MOSI	Output	SPI Master Output Slave Input (MOSI) signal for communication with SPI peripheral devices
11	SPI_MISO	Input	SPI chip/device select signal for communication with SPI peripheral devices
12	SPI_CS	Output	SPI Master Input Slave Output (MISO) signal for communication with SPI peripheral devices
13	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
14	GPIO1	Output	General-purpose software-programmable digital output port, 3.3V TTL level
15	GPIO2	Output	General-purpose software-programmable digital output port, 3.3V TTL level
16	GPIO3	Output	General-purpose software-programmable digital output port, 3.3V TTL level
17	GPIO4	Output	General-purpose software-programmable digital output port, 3.3V TTL level
18	GPIO5	Output	General-purpose software-programmable digital output port, 3.3V

			TTL level
19	GPIO6	Output	General-purpose software-programmable digital output port, 3.3V TTL level
20	GPIO7	Output	General-purpose software-programmable digital output port, 3.3V TTL level

4.6.ATP2500

The module utilizes a dual-row 12-pin fenced header and a USB Type-C interface for system power supply and communication. The specific pin descriptions are as follows:

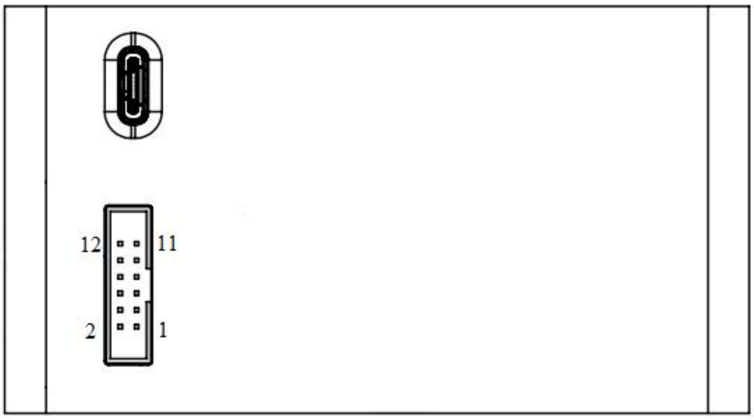


Table5.6 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	VCC	/	Power, 5V±0.5V
2	GND	/	Ground
3	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level
4	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level
5	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
6	Analog Out (0-5V)	Input /Output	12-bit programmable analog output, 0-5V voltage range
7	MCU_SPI_MOSI	Output	SPI Master Output Slave Input (MOSI) signal for communication with SPI peripheral devices
8	MCU_SPI_MISO	Input	SPI Master Input Slave Output (MISO) signal for communication with SPI peripheral devices
9	MCU_SPI_SCK	Output	SPI clock signal for communication with SPI peripheral devices
10	MCU_SPI_CS	Output	SPI chip/device select signal for communication with SPI peripheral devices
11	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
12	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin, output, 5V TTL level

4.7.ATP333X/ATP533X

The module utilizes a dual-row 20-pin fenced header and a USB Type-C interface for system power supply and communication. The specific pin descriptions are as follows:

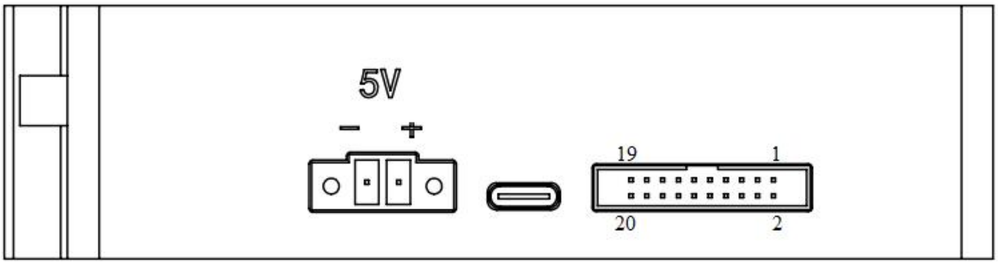


Table5.7 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	VCC	/	Power, 5V±0.5V
2	VCC	/	Power, 5V±0.5V
3	GND	/	Ground
4	GND	/	Ground
5	LD_RX	Input	TTL-level serial data receive pin for controlling the laser
6	LD_TX	Output	TTL-level serial data transmit pin for controlling the laser
7	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin, output, 5V TTL level
8	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
9	GPIO1	Output	General-purpose software-programmable digital output port, 3.3V TTL level
10	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
11	GPIO3	Output	General-purpose software-programmable digital output port, 3.3V TTL level
12	GPIO2	Output	General-purpose software-programmable digital output port, 3.3V TTL level
13	SPI2_SCK	Output	SPI clock signal for communication with SPI peripheral devices
14	SPI2_CS	Output	SPI chip/device select signal for communication with SPI peripheral devices
15	SPI2_MISO	Output	SPI Master Input Slave Output (MISO) signal for communication with SPI peripheral devices
16	SPI2_MOSI	Output	SPI Master Output Slave Input (MOSI) signal for communication with SPI peripheral devices
17	GND	/	Ground
18	3.3V	/	Power supply, 3.3V±0.1V
19	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level

20	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level

4.8.ATP50XX/ATP3040

The module utilizes a dual-row 20-pin fenced header and a USB 2.0 Type-B interface for system power supply and communication. The specific pin descriptions are as follows:

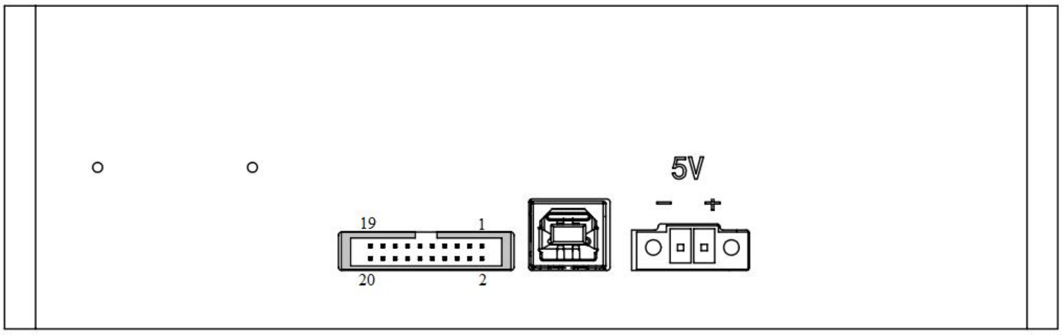


Table5.8 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	VCC	/	Power, 5V±0.5V
2	VCC	/	Power, 5V±0.5V
3	GND	/	Ground
4	GND	/	Ground
5	LD_TX	Output	TTL-level serial data transmit pin for controlling the laser
6	LD_RX	Input	TTL-level serial data receive pin for controlling the laser
7	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin, output, 5V TTL level
8	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
9	NC	/	/
10	NC	/	/
11	GPIO1	Output	General-purpose software-programmable digital output port, 3.3V TTL level
12	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
13	GPIO3	Output	General-purpose software-programmable digital output port, 3.3V TTL level
14	GPIO2	Output	General-purpose software-programmable digital output port, 3.3V TTL level
15	GPIO5	Output	General-purpose software-programmable digital output port, 3.3V TTL level
16	GPIO4	Output	General-purpose software-programmable digital output port, 3.3V TTL level
17	GND	/	Ground
18	+3.3V	/	Power supply, 3.3V±0.1V

19	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level
20	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level

4.9.AT5020P/ATP6500

The module utilizes a dual-row 20-pin fenced header and a USB Type-C interface for system power supply and communication. The specific pin descriptions are as follows:

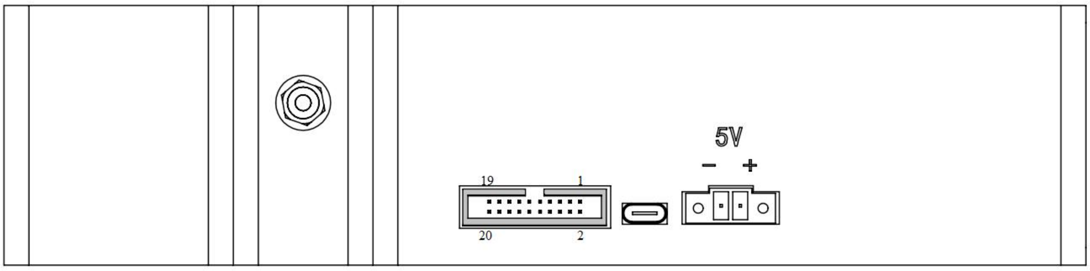


Table5.9 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	VCC	/	Power, 5V±0.5V
2	VCC	/	Power, 5V±0.5V
3	GND	/	Ground
4	GND	/	Ground
5	LD_TX	Output	TTL-level serial data transmit pin for controlling the laser
6	LD_RX	Input	TTL-level serial data receive pin for controlling the laser
7	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin, output, 5V TTL level
8	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
9	NC	/	/
10	NC	/	/
11	GPIO1	Output	General-purpose software-programmable digital output port, 3.3V TTL level
12	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
13	GPIO3	Output	General-purpose software-programmable digital output port, 3.3V TTL level
14	GPIO2	Output	General-purpose software-programmable digital output port, 3.3V TTL level
15	GPIO5	Output	General-purpose software-programmable digital output port, 3.3V TTL level
16	GPIO4	Output	General-purpose software-programmable digital output port, 3.3V TTL level
17	GND	/	Ground
18	+3.3V	Output	Power supply, 3.3V±0.1V
19	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level

20	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level

4.10.ATP8000/ATP8200

The module utilizes a dual-row 20-pin fenced header and a USB 2.0 Type-B interface for system power supply and communication. The specific pin descriptions are as follows:

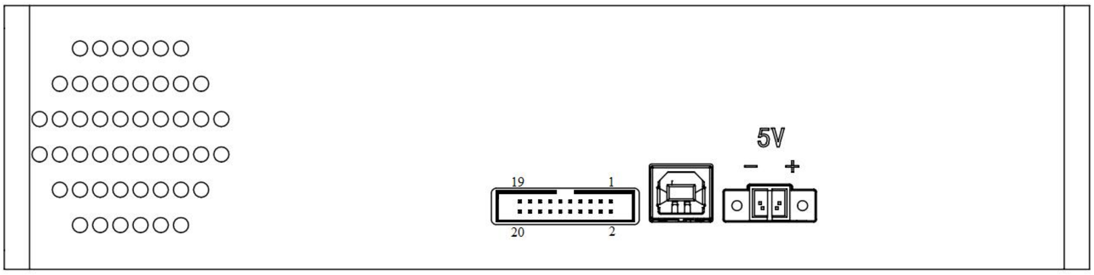


Table5.10 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	VCC	/	Power, 5V±0.5V
2	VCC	/	Power, 5V±0.5V
3	GND	/	Ground
4	GND	/	Ground
5	LD_TX	Output	TTL-level serial data transmit pin for controlling the laser
6	LD_RX	Input	TTL-level serial data receive pin for controlling the laser
7	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin, output, 5V TTL level
8	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
9	NC	/	/
10	NC	/	/
11	GPIO1	Output	General-purpose software-programmable digital output port, 3.3V TTL level
12	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
13	GPIO3	Output	General-purpose software-programmable digital output port, 3.3V TTL level
14	GPIO2	Output	General-purpose software-programmable digital output port, 3.3V TTL level
15	GPIO5	Output	General-purpose software-programmable digital output port, 3.3V TTL level
16	GPIO4	Output	General-purpose software-programmable digital output port, 3.3V TTL level
17	GND	/	Ground
18	+3.3V	/	Power supply, 3.3V±0.1V
19	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level
20	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level

4.11.ATP8600

The module utilizes a dual-row 10-pin fenced header and a USB Type-C interface for system power supply and communication. The specific pin descriptions are as follows:

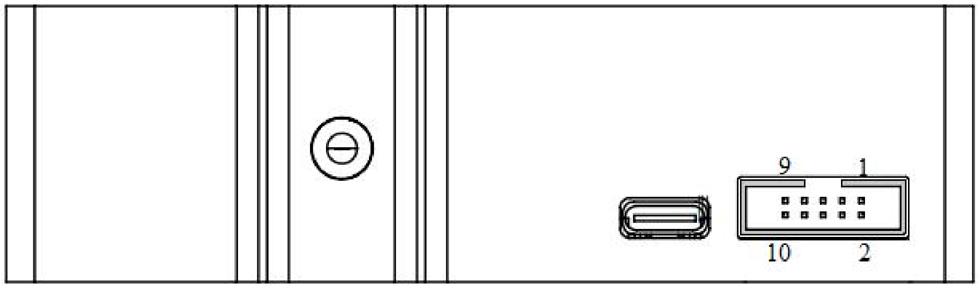


Table5.11 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level
2	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level
3	GPIO1	Output	General-purpose software-programmable digital output port, 3.3V TTL level
4	GPIO2	Output	General-purpose software-programmable digital output port, 3.3V TTL level
5	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
6	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
7	GND	/	Ground
8	GND	/	Ground
9	VCC	/	Power, 5V±0.5V
10	VCC	/	Power, 5V±0.5V

4.12.ATP8730GC

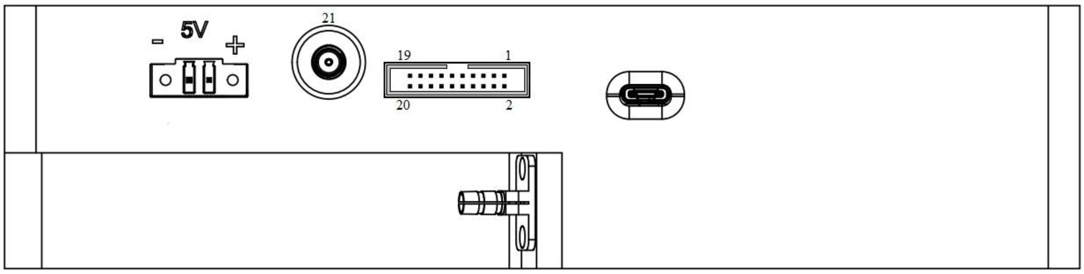


Table5.11 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	VCC	/	Power, 5V±0.5V
2	VCC	/	Power, 5V±0.5V
3	GND	/	Ground
4	GND	/	Ground
5	Lamp/LD Enable	Output	Light source (xenon lamp/laser) control pin, output, 5V TTL level
6	Single_strobe	Output	General-purpose software-programmable digital output port, 3.3V TTL level
7	SPI_SCK	Output	SPI clock signal for communication with SPI peripheral devices
8	SPI_MOSI	Output	SPI Master Output Slave Input (MOSI) signal for communication with SPI peripheral devices
9	SPI_CS	Output	SPI Master Input Slave Output (MISO) signal for communication with SPI peripheral devices
10	SPI_MISO	Input	SPI chip/device select signal for communication with SPI peripheral devices
11	GPIO0	Output	General-purpose software-programmable digital output port, 3.3V TTL level
12	GPIO1	Output	General-purpose software-programmable digital output port, 3.3V TTL level
13	GPIO2	Output	General-purpose software-programmable digital output port, 3.3V TTL level
14	GPIO3	Output	General-purpose software-programmable digital output port, 3.3V TTL level
15	GPIO4	Output	General-purpose software-programmable digital output port, 3.3V TTL level
16	GPIO5	Output	General-purpose software-programmable digital output port, 3.3V TTL level
17	3.3V	/	Power supply, 3.3V±0.1V
18	GND	/	Ground
19	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level

20	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level
21	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered

4.13.ATP8130

The module utilizes a dual-row 10-pin fenced header and a USB Type-C interface for system power supply and communication. The specific pin descriptions are as follows:

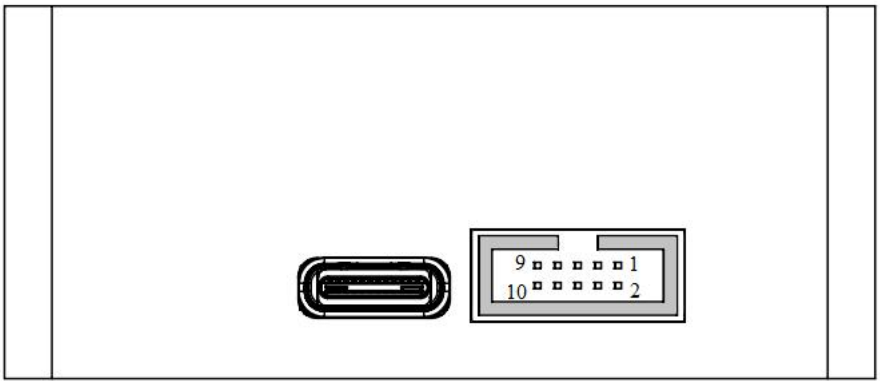


Table5.13 Electrical Pins

Pin	Pin Definitions	I/O	Description
1	GND	/	Ground
2	VCC	. ... /	Power Supply Voltage, 5V±0.5V
3	SPI_MOSI	Output	SPI Master Output Slave Input (MOSI) signal for communication with SPI peripheral devices
4	SPI_SCK	Output	SPI clock signal for communication with SPI peripheral devices
5	SPI_MISO	Input	SPI Master Input Slave Output (MISO) signal for communication with SPI peripheral devices
6	SPI1_NSS	Output	SPI chip/device select signal for communication with SPI peripheral devices
7	NC	/	/
8	Ext_trigger_in	Input	External trigger pin input, 3.3V TTL level, falling edge triggered
9	TTL_TX	Output	Serial data transmission pin with TTL level, 3.3V TTL level
10	TTL_RX	Input	Serial data reception pin with TTL level, 3.3V TTL level

Revision History

2017/02/13 V1.0
Initial version.

2017/06/28 V1.1
Modified description for reading wavelength calibration data.

2018/08/09 V1.2
Added xenon lamp pulse signal delay time.

2018/09/25 V1.3
Added GPIO0~GPIO11 pin control.

2019/02/26 V1.4
Added function to collect spectral data with xenon lamp turned off.

2020/03/18 V1.5
Added electrical interface pin descriptions.

2020/05/12 V1.6
Added external trigger function.

2020/11/12 V1.7
Added ATP1010 pin information.

2022/05/31 V1.8
Added SPI communication parameters.

2023/08/07 V1.9
Added spectrometer series pin descriptions.

2023/09/11 V2.0
Added 30-series pin descriptions.

2023/12/13 V2.1
Corrected definition errors for some spectrometers.

2023/12/25 V2.2
Corrected instructions.

2024/01/02 V2.2.1
Added Chinese version.

2024/01/24 V2.3

Added external trigger for ATP8600.

2024/10/16 V2.4.0

Version number updated;
Modified overall document format;
Added xenon lamp functionality for ATP series fiber spectrometers;
Added Modbus protocol for ATP series fiber spectrometers.

2024/11/19 V2.4.1

Fixed description errors.

2024/12/05 V2.4.2

Added ATP8730GC electrical interface pin descriptions.

2025/04/29 V2.4.3

Fixed erroneous descriptions in the Modbus protocol.

2025/05/23 V2.4.4

Added ATP8130 electrical interface pin descriptions;
Modified ATP1030 electrical interface pin descriptions.

2025/08/21 V2.4.5

Modified overall format of the general protocol;
Unified pin order in the electrical interface section.